

1. If $\mathbf{u}$ and $\mathbf{v}$ are two vectors in $\mathbb{R}^n$ with the same magnitude, then magnitude of $\mathbf{u} - \mathbf{v}$ must be zero. Is the above statement True or False? Justify.

2. Can two orthogonal vectors be a part of the same space?

3. If $\mathbf{u} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$ is orthogonal to vectors $\mathbf{v}$ and $\mathbf{w}$, does that mean $\mathbf{v}$ and $\mathbf{w}$ are parallel?

Explain.

4. Suppose $\mathbf{u}$ is orthogonal to vectors $\mathbf{v}$ and $\mathbf{w}$. Is $\mathbf{u}$ orthogonal to $2\mathbf{v} - 5\mathbf{w}$? Justify your answer.

5. Suppose $\mathbf{u}$ is orthogonal to $(\mathbf{v} + \mathbf{w})$ and $(\mathbf{v} - \mathbf{w})$. Check if $\mathbf{u}$ is orthogonal to $\mathbf{v}$.

6. Suppose $\mathbf{u}$ is orthogonal to vectors $\mathbf{v}$ and $\mathbf{w}$. Prove that $\mathbf{u}$ is orthogonal to all linear combinations of $\mathbf{v}$ and $\mathbf{w}$.

7. Suppose $\mathbf{u}$ is orthogonal to vector $\mathbf{v}$ and $\mathbf{u} \cdot \mathbf{w} = 3$. For what value of c is $\mathbf{u}$ orthogonal to $(\mathbf{v} + c\mathbf{w})$?

8. Suppose $\mathbf{u}$ is orthogonal to vector $\mathbf{v}$ and $\mathbf{w} = 3\mathbf{v} - 2\mathbf{u}$. Is $\mathbf{w}$ necessarily orthogonal to $\mathbf{u}$? Justify your answer.